

Assignment 6

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("C:/Users/Admin/Downloads/NEW DSBDA/iris.csv")
```

```
df.head()
```

```
df.info()
```

```
df.isnull().sum()
```

```
df.columns
```

```
X = df.drop(['Id', 'Species'], axis=1)
```

```
Y = df['Species']
```

```
xtrain, xtest, ytrain, ytest = train_test_split( X,Y,test_size=0.2,random_state=42)
```

```
gaussian = GaussianNB()
```

```
gaussian.fit(xtrain, ytrain)
```

```
GaussianNB()
```

```
Y_pred = gaussian.predict(xtest)
```

```
print(Y_pred)
```

```
cm = confusion_matrix(ytest, Y_pred)
```

```
print(cm)
```

```
accuracy = accuracy_score(ytest, Y_pred)
```

```
print("Accuracy =", accuracy)
```

```
precision = precision_score(
```

```
    ytest,
```

```
    Y_pred,
```

```
    average='macro'
```

```
)
```

```
print("Precision =", precision)
```

```
recall = recall_score(  
    ytest,  
    Y_pred,  
    average='macro'  
)  
print("Recall =", recall)
```